

University of South Florida
Industrial & Management Systems Engineering
EIN 4933/6935 Python for Data Science
Summer 2023

Time & place	Remote, 6:00PM – 10:15PM Thursday.
Instructor	Payman Ghasemi, Ph.D (c)., payman@usf.edu; office hours: by appointment.
TA	Ashim Khanal, ashimkhanal@usf.edu; office hours: by appointment.
Prereqs	Coding and stats experience (though it is not essential)
Grade scale	A = [90, 100], B = [80, 90), C = [70, 80), D = [60, 70), F = [0, 60) (<i>no +/- grades</i>)

All contents (even the timeline) in this document can possibly change during the semester.

1 Description

Python is a high-level, general-purpose programming language and free software environment. In recent years, its popularity increased significantly in academic and industry fields due to its ease of accessibility to a wide variety of packages. Students taking this course will learn how to program with python as well as applications to data science through its popular packages. The course materials include the lecture handouts and practice codes. In the first half of the course, Python programming topics (e.g., object-oriented programming, functional programming, data structures, environments, etc.) will be covered. The second half of the course will focus on data science applications. Several major Python data science packages will be introduced.

2 Objectives

The objective of the course is to provide hands-on experience for students who desire to use Python to develop data analytics/Science applications. At the end of this course, students will be able to:

- Write Python code in object oriented and functional programming fashions
- Use available packages in Python for data analytics
- Be familiar with the different types of data structures in Python
- Manipulate large amounts of data by using Python libraries
- Perform and interpret predictive analytics (regression and classification) by using classic and machine learning algorithms
- Use data visualization tools

3 Policies

- The deadlines are STRICT, and late submissions will NOT be accepted. The LAST time that you can submit the requested materials is at 11:59 p.m. on the specified dates. All submissions MUST be done electronically in Canvas.
- Any violation of academic honesty will be dispatched in accordance with the university policies.
- Students in need of academic accommodations for a disability need to contact the SAS Office.
- Students are encouraged to collaborate with their classmates. However, they must independently submit their homework assignment.
- For other applicable policies, please refer to the current version of the undergraduate catalog.

4 Course Requirements

- Assignments (70%): Homework assignments will allow students to improve their programming skills in Python. In addition, students will be tasked to use specific functions from a package. The main goal is that students implement their understanding of the concepts covered during the lecture. The tentative number of homework assignments is 7 (it might be more or less, though).
- Final Project (30%): A team of four students (randomly selected) will determine a data science project, build a Python notebook and show the results with visuals. Students must consult with the instructor to confirm the project topic with a paragraph description. Deliverables of the project will consist of a presentation with 5 slides (problem description, data summary, data visualization, model performance, result visualization) and a Python notebook.

5 Schedule

Weeks	Date	Topic	Lecture
1	5/18	Introduction	Welcome Intro to Anaconda Install libraries Creating packages with Python
2	5/25	Data structures	Arrays, matrices, tuples, lists, data frames
3-4	6/1 6/8	Programming basics	Control flow and loops Functions and environments Python object oriented programming Functional programming
5	6/15	Data manipulation	Data import/export Dataframe manipulation(join, concat, merge,...)
6	6/22	Visualization	Plotting and visuals Python Markdown
7-10	6/29 7/6 7/13 7/20	Data Science	Training models Classic regression models Machine learning algorithms Decision trees, random forest Feature engineering algorithms: Feature elimination and feature extraction algorithms Correlation analysis, Recursive feature elimination Principle component analysis K-means clustering Hierarchical Clustering